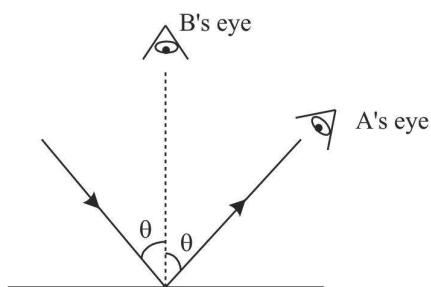


Hints & Solutions

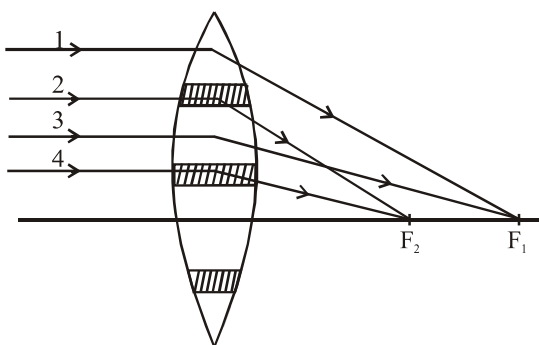
1. When the scattering particles are of size smaller than the wavelength of light then only Rayleigh's law is applicable. Due to this traffic signals are of red colour.
2. Paraxial rays are those rays which are incident at a point close to the pole of the mirror and makes small angle with principal axis
3. By taking the laws of reflection to be true for all points of the total mirror, the image will be that of the whole object but as the area of reflecting surface has been reduced, the intensity of image will decreases.
4. Optical fibre has core and cladding in which the refractive index of core is greater than refractive index of cladding so that total internal reflection takes place from the boundary of core-cladding interface.
5. Dispersion takes place because refractive index of medium for different frequency is different for eg $\rightarrow$ The bending of red light component is least and that of violet part is maximum
6. It is due to refraction of light through the atmosphere. By actual sunrise we mean the actual crossing of the horizon by the sun. Due to refraction the apparent shift in the direction of sun is about half a degree and corresponding time difference is about 2 minutes.
7. This is due to mirages which are caused by total internal reflection.
8. When sunlight is incident on the particle whose size is large as compared to wavelength then scattering of all colours are same.
9. optical aberration i.e. defects of image due to lenses i.e. chromatic and spherical aberration can be minimized by using multi component lenses.
10. When white light passes through thick lenses, red and blue colour are focussed at different points which is called chromatic aberration
11. Rainbow formation is due to the combined effect of dispersion, total internal reflection and refraction. When sunlight is incident on rainwater droplet then dispersion occurs from Ist face, internal reflection occurs from IInd face and again refraction occurs from Ist face by which rainbow is formed
12. An observer can see a rainbow only when his back is towards the sun i.e. sun should be shining in one part of sky and it should be raining in opposite part of the sky.
13. In primary rainbow there is in one internal reflection where as in secondary rainbow there are two internal reflection.
14. There is no chromatic aberration in mirror, so telescope with mirror objectives are called reflecting telescopes
15. Laws of reflection are valid for all types of surfaces i.e. plane and rough surface
16. Frequency does not change during refraction so the velocity of red colour is maximum in glass.
17. A passenger in an aeroplane may see primary and secondary rainbow as concentric circles.
18. Parabolic concave mirror is used because it is free from spherical and chromatic aberration.
20. vacuum is a non-dispersive medium in which all colours travel with same speed.
21. When person is between focus & centre of curvature and at focus, his inverted image is formed behind him, but when he stands beyond C, image is formed between focus and center of curvature is in front of him and thus he is able to see his image.
26. Because refractive index of lens material is greater than medium therefore air bubble show diverging nature
27. $M.P. = 1 + \frac{D}{f}$
 $\therefore f_v < f_r$
 so, the magnifying power is larger when the object is seen in violet light.
28. As the refracting angle is small deviation
 $= (\mu - 1) A$
 angle of dispersion
 $= (\mu_v - 1)A - (\mu_r - 1)A = (\mu_v - \mu_r)A$
 $= \left(\frac{\mu_v - \mu_r}{\mu_{\text{mean}} - 1} \right) A (\mu_{\text{mean}} - 1) = \omega A (\mu_{\text{mean}} - 1)$
 $= (0.053) \times 3 \times (1.68 - 1) = 0.108^\circ$
29. Cauchy's formula is $\mu = A + \frac{B}{\lambda^2}$

30. The light reflects of the coin and reaches A's eye.



As the polished half is smoother than the unpolished half, hence the polished half behaves more like a mirror. So, most of the light striking the polished half of the coin reflects instead of going diffusion reflection. Hence, A sees a high percentage of the light that hits the polished part of the coin. Light hitting the unpolished half undergoes more diffuse reflection, bouncing off the coin in all different directions. Hence, a smaller percentage of that light reaches A's eyes. So, for A the polished half looks more brighter than the unpolished half.

31. Image can be formed on the screen if it is real. Real image of reduced size can be formed by a concave mirror or a convex lens of maximum focal D/4.
32. Shown in the ray diagram, is an incident beam of parallel rays 1,2,3 & 4. Rays 1 & 3 pass through the medium 1



$\Rightarrow$ It has a focal length f_1 given as

$$f_1 = \frac{R_1}{2(n_1 - 1)}$$

Similarly rays 2 and 4 passes through the medium 2.

$\Rightarrow$ It has a focal length

$$f_2 = \frac{R_1}{2(n_2 - 1)}$$

when $n_2 > n_1$, $f_2 < f_1$ and vice versa.

Therefore, there are two focal points at right

hand sides; for the other sides $f'_1 = \frac{R_2}{2(n_1 - 1)}$

and $f'_2 = \frac{R_2}{2(n_2 - 1)}$. Therefore, another two

focal points are for left hand side. We see that the focal points (focal lengths) need not to be same if $R_1 \neq R_2$ we obtain four focal points but they are different from each other.

33. Alexander's dark band lies between primary and secondary rainbows, forms due to light scattered into this region interfere destructively. Primary rainbow subtends an angle nearly 41° to 42° at observer's eye, whereas secondary rainbows subtends an angle nearly 51° to 54° at observer's eye w.r.t. incident light ray.

So the scattered rays with respect to the incident light of the sun lies between approximately 42° and 50° .